

Plastic Alternatives: Exploring Mycelium as a Medium

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Abstract

This paper discusses using mycelium, the vegetative part of a fungus, as a medium for (1) degrading plastic and (2) growing alternative and sustainable materials—both of which address the rise in global plastic consumption and pollution. A series of prototypes were made using Ecovative's *Reactivating Dry Material* to demonstrate the potential uses for these mycelium-grown products and to show how easily we can substitute such materials for those that are traditionally made from polyurethane and other harmful, toxic plastics. In addition, preliminary experiments were conducted to understand the potential for bioremediation using the fungus organism *schizophyllum commune*. Using both yeast malt and potato dextrose agar mediums, ten plastic samples were examined for 15 days. The yeast malt agar mediums yielded the best results. In less than a week the fungus samples on yeast malt plates grew to approximately twice the size of those on the potato dextrose plates. The 10 samples continue to be under observation and will be observed until degradation is complete and the plastic is gone. As of this writing, the *coffee cup lid* sample appears to already be undergoing degradation. Further research on timing using a broader selection of polyurethane samples will be completed over the next 12 months as part of the New Challenge Sustainable Design Award.

Keywords: mycelium, *schizophyllum commune*, bioremediation, grown materials, Ecovative, fungi, sustainable design, plastic degradation, bio design.

Plastic: Environmental Climate & Cultural Context

The Plastics Industry is the third largest manufacturing industry in the United States contributing billions of dollars to this country's economy (18). According to a recent report, "global production of plastics grew from around 1.5 million tons in 1950 to 280 million tons in 2012" (19). Of the plastic produced globally, the packaging industry is the largest "end-user," constituting nearly 40% of the entire plastics market. Of the 25.1 million tones of plastic that is collected from post-consumer waste, nearly 65% of that is from packaging (17). Figure 1 shows the breakdown of post-consumer waste by sector. The global production and waste of plastics is staggering, and just the sheer size alone has environmental implications.

As an example of the global impact and harmful effects of plastics, consider that according to a European report, nearly 4.7 tons of plastic is dumped into the ocean annually (4). The Great Pacific Garbage Patch is an area in the Pacific Ocean where a massive accumulation of debris occurs because of ocean currents such as the North Pacific Subtropical Gyre (15). This accumulation of

debris is mainly due to non-biodegradable plastics that never fully break down; rather they break into smaller pieces called micro-plastics. These micro-plastics are extremely harmful to the ocean because they collect into a mass or a "cloudy soup" that floats on the water's surface (covering it for miles) and prevent sunlight from entering the water and providing nutrients to the algae and plankton below (15).

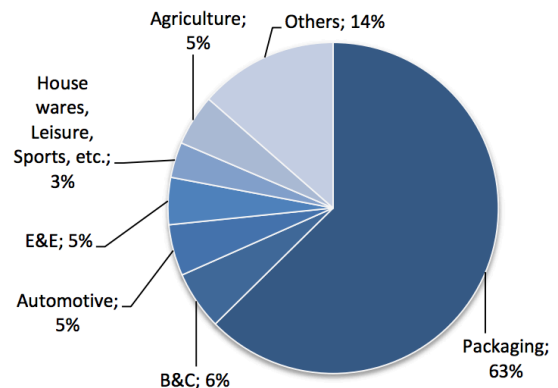


Figure 1: Generation of post-consumer plastics waste by application, 2011 Figure courtesy of (17)

It is almost impossible to avoid plastic today. It is all around us, covering most of our products and environments. It has proven to be one of the most toxic and harmful products to come out of the 21st century industrial boom. Happily however, nature has provided us with a far superior material: mycelium. Mycelium is not only non-toxic and virtually waste-free, but it can teach us about the intricate connection between the environment and ourselves.

Precedents: Mycelium as Alternative Material

For the better part of the last decade, the design and architecture world has been experimenting with the use of mycelium as an alternative material as a means of considering sustainable design as best practice. As an example, at Material ConneXion the company Ecovative shows six different product lines, from packaging to insulation, that use mycelium as a material. A design using an Ecovative building product was awarded the 2014 Young Architect Award at MoMA PS1. The firm The Living won the award for their project *Hy-Fi* which "use[d] biological technologies combined with cutting-edge computation and engineering to create new building materials, a new method of bio-design, and a structure...that is 100% grown and 100%

compostable” (1). The project is currently on display in the Architecture and Design section at MoMA. Like Ecovative, MycoWorks is a biomaterials development company that creates products that use mycelium as a material. In 2012 MycoWorks co-founder Philip Ross designed a line of mycelium-made furniture that is currently on display at San Francisco’s Exploratorium museum of science (21).

Both MycoWorks and Ecovative are companies at the forefront of innovation; they are reimagining our world with products that are “grown” rather than “machined.” Making these materials available to designers and architects is allowing for an *ecology* revolution where the harmful, toxic, wasteful industrial processes of the 20th century are being supplanted by green, virtually *waste-less* systems. This may be the key to the future of sustainable design and construction. But how do we address our past and the damage that has already been done? How do we dispose of all that plastic?

Mycelium: Material for De-constructing Environments

Mycoremediation is a soil decontamination technique that uses mycelium developed by Peter Stamets, a world-renowned mycologist. He has famously said in a TedTalk that “mycelium are the grand molecular *dissemblers* of nature” (24).

Mycelium are particularly good at “breaking down large molecules into smaller molecules that can then be used by other members in the ecological community” (23) and thus are prime candidates for bioremediation—a treatment that uses naturally occurring organisms to break down hazardous substances into less toxic or non-toxic substances.

New Mycelium Discoveries: *Pestalotiopsis Microspora*

In 2011 a team from Yale University’s Department of Molecular Biophysics and Biochemistry travelled to the Yasuni National Forest in the Ecuadorian Amazonian rainforest. It was on that trip that they made a revolutionary discovery: *Pestalotiopsis Microspora*. In their subsequent paper *Biodegradation of Polyester Polyurethane by Endophytic Fungi* they described how two isolates of this endophytic fungi organism were able to degrade the synthetic polymer polyester polyurethane (PUR) (22). Figure 2 shows an infrared spectrum of a polyurethane sample before and after the fungus was used. After six days of incubation with the fungus *Pestalotiopsis microspora*, the PUR peak disappears after degradation of fungi (22).

Their finding has profound implications for plastic bioremediation. Unfortunately, while discussing this paper and work with Professor Scott Strobel at Yale, the lead faculty advisor of the main paper (22) and original discovery, he mentioned that at this time the organism *Pestalotiopsis microspora* is not available for use outside of his lab due to regulations by the Ecuadorian government.

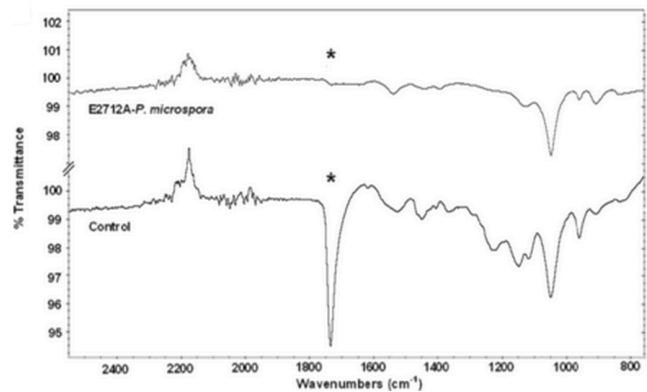


Figure 2: Infrared spectra of PUR liquid medium – sample spectrum (top) and control spectrum (bottom). Russel, et al (22)

Bio Design: Implications for Design

In 2012 MoMA, New York’s Museum of Modern Art, published a book called *BioDesign: Nature + Science + Creativity*, which was perhaps the first time the term *BioDesign* was used. *BioDesign* refers not just to design that is “biology-inspired” but to design that incorporates “living organisms as essential components that enhance the function of the finished work” (14).

Microbial Home by Philips Design in the Ecological Object Engineering section of the book describes a future home where all the home appliances work together like an ecosystem. One imagined appliance is called the *Paternoster*—a conceptual device that would “up-cycle plastics.” This device would allow the plastic we consume in our home to be “ground into small pieces that can be digested naturally by fungi, which can, in turn, be harvested and eaten” (14). Philips describes this design as a “far future” concept that is “not intended as a production prototype” (16).

However, the design firm *Living Studios*, taking inspiration from the paper that came from the Yale team, created a design concept called *Fungi Mutarium: Growing Food On Toxic Waste*. In collaboration with Utrecht University’s Prof. Dr. Han Wösten and Kasia Lukasiewicz, they worked with two fungi organisms (*Schizophyllum Commune* and *Pleurotus Ostreatus*) to biodegrade plastic and produce edible byproducts (See Figures 3 and 4). Although this project is also a conceptual device that is not yet commercially viable, the lab-based research and experimentation is pushing the Yale research into the GIY and bio-hacking realm where further investigation and prototyping can be explored.

BioDesign is an inspiring book that imagines how all aspects of our lives can be reimagined using biology as the core of a design in order to work symbiotically with nature rather than opposed to it.

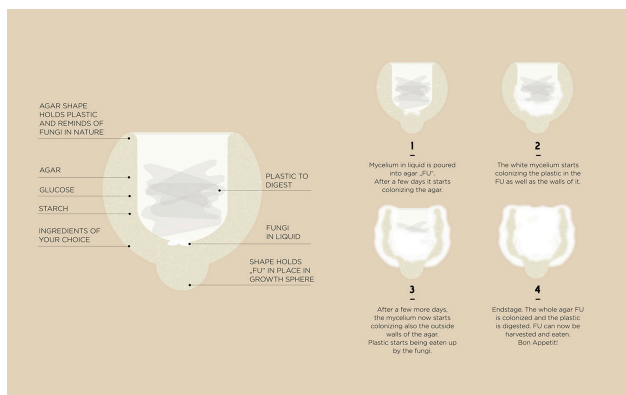


Figure 3: *Fungi Mutarium* by Livin Studio. Diagram of the growth process.



Figure 5: *Fungi Mutarium* by Livin Studio. Image of the complete system

PUR BIO REM

PUR BIO REM (Polyurethane Bioremediation) is a project that picks up where *Fungi Mutarium* left off by pushing the design process further to develop a full-fledged commercial product that can be deployed for mass use. *PUR BIO REM* won the 2015 New Challenge Award for sustainable design (one of ten winners) and over the course of the 2015-2016 academic year will be awarded funding to implement the proposal (3).

The *PUR BIO REM* team has developed a recycling system inside a standard trashcan. The system utilizes fungi that have the ability to degrade synthetic polymer polyester polyurethane (PUR), which is disposed of inside the trashcan. The team will construct these state-of-the-art waste receptacles to have an inner cavity where the fungi will live and break down the plastic waste.

The project serves as a revolutionary method in plastic waste management and re-imagines “recycling” by involving nature in the process. Most importantly, it serves to help users think about their plastic consumption.

PUR BIO REM: Trash Can Mechanics

Figure 5 shows a mock-up of what the *PUR BIO REM* system will look like. A preliminary design (before user testing and

further research into the mechanics of the fungi) involves three components. First, users will place plastic items in the receptacle. The plastic will pass through a mechanical device that will chop the plastic into smaller pieces. Second, the plastic will then be deposited onto the fungi plates where decomposition will occur. Third, once the plastic is completely digested, the fungi will be removed and transferred to a composting site where it will provide rich nutrients for the soil and further aid decomposition on organic material.

The experiments described in the next section of this paper were performed as the initial “proof of concept” for the first phase of prototypes for *PUR BIO REM*.



Figure 4: *PUR BIO REM* concept image

Primary Prototypes: Experiments in Bioremediation with *Schizophyllum Commune*

Schizophyllum commune is the world’s most widely distributed fungus, occurring on every continent except Antarctica. It is a ubiquitous white rot fungus and generally considered to be a good candidate for bioremediation due to its ability to degrade complex plant biomass (10).

Lab Notes

It was not possible to procure *Pleurotus Ostreatus* samples as this particular organism is not available from the biological supply company Carolina (5). Further investigation as to how to procure the sample will need to be made for further research. Samples of *schizophyllum commune* were procured from Carolina (5).

Because of limited access to biological laboratory facilities all experiments were conducted in the private residence of the author. As a result, the potential for sample contamination must be considered when examining the results.

The experiments were performed as follows:

1. Using gloves and 70% rubbing alcohol, the surfaces of the work area, bottles, dishes, pans, etc. were sanitized.
2. Two agar mediums were prepared.
 - Two tablespoons of potato dextrose agar were

combined with one tablespoon of water, boiled under low heat, cooled, and poured into four petri dishes, covering the bottom with a thin layer of agar.

- A glass bottle of solid yeast malt agar was placed in a pot with boiling water under low heat until the agar melted, the liquid cooled, and then poured into six petri dishes, covering the bottom with a thin layer of agar.
3. Ten different “tests” were performed using a variety of the type of plastics commonly found in the home thereby potentially simulating the types of plastics consumers may use. Table 1 shows the different plastics tested. Each sample was placed in a petri dish.
 4. Once the agars had cooled (approximately 30 mins from pouring), the agar mediums were ready for use.
 5. The fungi samples come in a glass jar. Using a scalpel, a small piece of the organism was transferred to each petri dish.
 6. Each dish was labeled with the agar type and plastic sample.
 7. The dishes were set aside in a dark room to grow.

Table 1: Results from *schizophyllum commune* experiments after 15 days of observation

Plastic Type	Specimen Performance Observations
Potato Dextrose Agar	
Dry Cleaning Bag	No fungal growth
Garbage Bag	No fungal growth
Random Plastic Bag	No fungal growth
Raman Noddle Packaging	No fungal growth
Yeast Malt Agar	
Frozen Fruit Packaging	Significant fungal growth; plastic bag appears to be intact. No degradation.
Garbage Bag	Significant fungal growth. Bag appears discolored and is being engulfed by the fungi but seems to still be in tact.
Garbage Bag Tie	Moderate fungal growth. This sample had the least growth of all the yeast malt samples. The edges of the sample are discolored and appear to be breaking down, but this could just be an effect of sitting on the agar.
Dry Cleaning Bag	Significant fungal growth; however, the fungi has grown around the bag and not over it. Degradation does not appear to be taking place.
Dry Cleaning Collar Stiffener	Significant fungal growth. Same observation as the Dry Cleaning Bag. The fungi have grown around the plastic and do not seem to be degrading it.
Coffee Cup Lid	Significant fungal growth. This sample appears to have the best results. The fungi have grown over the top of the plastic sample. The plastic sample does not appear to have changed in color, but the envelopment of the sample by the fungi is a promising result.

As of the time this paper was written, the fungi had been growing for 15 days. Yeast malt agar as a medium for fungal growth yielded the best results. In less than a week the samples on a yeast malt plate has grown approximately twice their size while the

potato dextrose plates did not grow at all. The *coffee cup lid* appears to be experiencing the best results. Figure 6 shows the samples after 15 days. All of the 10 samples will continue to be under observation for the next three months to see if degradation occurs. If degradation does not occur, certain factors such as environmental conditions and the organism strain may need to be modified for the next phase of the experiment.



Figure 6: Image of the 10 plastic samples undergoing bioremediation from *schizophyllum commune* after 15 days incubation

Further Bioremediation Research & Next Steps

The points below outline the next steps to further research into the use of fungi to degrade plastic:

- Access to a full-scale biology lab may help improve results by minimizing contamination.
- Further research should be considered to examine bacterial organisms that have plastic degradation properties. Two experiments should be considered:
 - The potential for bacteria to operate on their own to degrade plastic and
 - The potential for fungal organisms to work in partnership with the bacterial organisms.
- Further communications should be initiated with Professor Scott Strobel at Yale. Access to the *Pestalotiopsis Microspora* could allow for a more robust study.
- *Pleurotus Ostreatus* should be considered as an alternate organism for study. Procurement may need to be through a lab facility as Carolina does not provide this organism.
- Finally, a more robust and long-term study on the *schizophyllum commune* organism needs to be performed:

1. Contact plastic manufactures to procure liquid-form polymer polyester polyurethane (PUR).
2. Understand the types of plastic (chemical makeup) to study effects.
3. Design a study to test many more types of plastics, i.e. 50 plates.
4. Test the fungi response in different environmental conditions, i.e. no light, outside vs. inside.

Supplemental Experiments with Ecovative's GIY

Ecovative in particular is fulfilling their mission to “rid the world of toxic, unsustainable materials” by championing the GIY movement. Just as the DIY movement has revolutionized open-source physical computing and hardware, the *Grow It Yourself* movement is making biology open-source and has even started a bio-hacking subculture. Ecovative sells mushroom material kits allowing everyone to experiment in growing their own products.

Using the Ecovative GIY kit, a number of prototypes were made as proof of concept that anyone can grow their own materials with 0% pollution, 0% waste and at a fraction of the cost of non-sustainable materials such as plastics. Table 2 is a breakdown of the possible uses for the future mycelium products in lieu of the more traditional environmentally harmful or wasteful ones. The three of particular concern are the ones that use plastic products such as Styrofoam, Polyurethane foam and Polyvinyl Chloride (PVC).

Table 2: Prototypes to supplant traditional non-sustainable materials

<i>Product Prototype</i>	<i>Traditional Material</i>
Building Material - Brick	Clay aggregate or concrete
Building Material - Dry Wall	Plaster
Packing Peanuts Shipping Material	Styrofoam
Perforated Packing Material	Polyurethane Foam
Corrugated plastic sheet	Polyvinyl chloride (PVC)

The process for making the prototypes is described in Figure 7 which shows an infographic of the process.

Figures 8 – 10 show the three prototypes made to address these harmful plastics. For further details on this experiment and to see all the photos, refer to the blog post *Mycelium Experiments: Growing Building Materials* (9).



Figure 7: Prototype of flat mycelium sheet that could be used in place of PVC materials



Figure 8: Prototype of corrugated mycelium sheet that could be used in place of Polyurethane Foam used for packaging



Figure 9: Prototype of small mycelium balls that could be used in place of Styrofoam “popcorn” used for packaging

10 The Future Imagined through BioDesign

The experiments conducted in this paper, in both bioremediation and mycelium as a medium to replace traditional plastic packaging, are both prototypes for a new way of acting in the world. They showcase this new school of thought, *BioDesign*—a revolutionary way of operating in the world that supplants the arcane, industrial complex of the 20th century that has caused unquantifiable damage to our planet. This new modality envisions biology as the new “building” material for all aspects of modern life. “Bio design is not about merely taking cues from organic structures and operations. It's about harnessing the machinery of

the natural world to perform as nature does: storing and converting energy, producing oxygen, neutralizing poisons and disposing of waste in life-sustaining ways..." (12). The future of our world depends on our ability to realize prototypes like those described in this paper. Mycelium is the medium of the future, and the sooner we begin implementing these powerful organisms into all facets of design thinking, the better off we will be tomorrow.

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